Register No.																

SRM Institute of Science and Technology
College of Engineering and Technology
School of Computing
(Common to all branches)

SET - A

Academic Year: 2023-24 (ODD)

Test: CLA-T2 **Date: 2-11-2023**
Course Code & Title: 21CSC203P Advanced Programming Practice **Duration:** 100 minutes
Year & Sem: II Year / III Sem **Max. Marks:** 50

Course Articulation Matrix: ***(to be placed)***

S.No.	Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	CO1	3	2	-	-	-	-	-	-	-	-	-	-	-	2	-
2	CO2	3	2	-	1	-	-	-	-	-	-	-	-	-	2	-
3	CO3	3	-	2	-	2	-	-	-	-	1	-	-	-	2	-
4	CO4	3	2	-	1	-	-	-	-	-	-	-	-	-	2	-
5	CO5	3	-	2	1	2	-	-	-	-	1	-	-	-	2	-

Part - A
(5 x4 = 20 Marks)
Instructions: Answer any 5 Questions

Q. No	Question	Marks	BL	CO	PO	PI Code
1	**What are the advantages of multi-threading?** **Ans:** 1. Responsiveness 2. Resource Sharing, 3. Message Passing ,Shared Memory 4. Economy 5. Scalability 6. Better Communication 7. Utilization of multiprocessor architecture 8. Minimized system resource usage **Note:** Explain list of advantages	4	L2	3	1	2.4.1

2	**Compare AWT and Swing.** **Ans:**		4	L4	3	2	1.4.1
	AWT	**Swing**					
	The AWT Component classes are provided by the java.awt package.	The Swing component classes a provided by the javax.swi package.					
	The Components used in AWT are mainly dependent on the operating system.	The Components used in Swi are not dependent on t operating system. It is complete scripted in Java.					
	The AWT is heavyweight since it uses the resources of the operating system.	The Swing is mostly lightweig since it doesn't need a Operating system object f processing. The Swi Components are built on the t of AWT.					
	The Appearance of AWT Components is mainly not configurable. It generally depends on the operating system's look and feels.	The Swing Components a configurable and mainly suppo pluggable look and feel.					
	The Java AWT provides a smaller number of components in comparison to Swing.	Java Swing provides a great number of components tha AWT, such as list, scroll pane tables, color choosers, etc.					
	Java AWT stands for Abstract Window Toolkit.	Java Swing is mainly referred as Java Foundation Classes (JFC					
	Java AWT has 21 peers. There is one peer for each control and one peer for the dialogue. Peers are provided by the operating system in the form of widgets themselves.	Java Swing has only one peer the form of OS's window obje which provides the drawi surface used to draw the Swing widgets (label, button, ent fields, etc.) developed directly Java Swing Package.					
	Java AWT many features that are completely developed by the developer. It serves as a thin layer of development on the top of the OS.	Swing components provide t higher-level inbuilt functions f the developer that facilitates t coder to write less code.					

<table>
<tr><td></td><td><table>
<tr><td>Java AWT needs a higher amount of memory for the execution.</td><td>Java Swing needs less memo space as compared to Java AWT</td></tr>
<tr><td>Java AWT is slower than swing in terms of performance.</td><td>Java Swing is faster than t AWT.</td></tr>
</table></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>3</td><td>Write a python cocde to print the even numbers, odd numbers from the list of given numbers using Imperative approach [2,5,8,23,14,37,47,18,20,36,29]

<u>Ans:</u>

num = [2,5,8,23,14,37,47,18,20,36,29]

odd =[]

even =[]

for i in num:

if i%2==0:

even.append(i)

else:

odd.append(i)

print(odd)

print(even)</td><td>4</td><td>L2</td><td>4</td><td>2</td><td>2.4.2</td></tr>
<tr><td>4</td><td>Differentiate TCP and UDP.

<u>Ans:</u>

<table>
<tr><th>TCP</th><th>UDP</th></tr>
<tr><td>It stands for Transmission Control Protocol.</td><td>It stands for User Datagran Protocol.</td></tr>
<tr><td>It is a connection-oriented protocol, which means that the connection needs to be established before the data is transmitted over the network.</td><td>It is a connectionless protocol which means that it sends th data without checking whethe the system is ready to receive o not.</td></tr>
<tr><td>TCP is a reliable protocol as it provides assurance for the delivery of data packets.</td><td>UDP is an unreliable protoco as it does not take the guarante for the delivery of packets.</td></tr>
</table></td><td>4</td><td>L2</td><td>4</td><td>2</td><td>2.2.1</td></tr>
</table>

	TCP is slower than UDP as it performs error checking, flow control, and provides assurance for the delivery of	UDP is faster than TCP as i does not guarantee the deliver of data packets.					
	The size of TCP is 20 bytes.	The size of the UDP is 8 bytes.					
	TCP uses the three-way-handshake concept. In this concept, if the sender receives the ACK, then the sender will send the data. TCP also has the ability to resend the lost data.	UDP does not wait for an acknowledgment; it just send the data.					
	It follows the flow control mechanism in which too many packets cannot be sent to the receiver at the same time.	This protocol follows no suc mechanism.					
	TCP performs error checking by using a checksum. When the data is corrected, then the data is retransmitted to the receiver.	It does not perform any erro checking, and also does no resend the lost data packets.					
	This protocol is mainly used where a secure and reliable communication process is required, like military services, web browsing, and e-mail.	This protocol is used where fas communication is required an does not care about th reliability like VoIP, gam streaming, video and musi streaming, etc.					
5	**What is Socket? Explain in details.** **Ans:** A socket is a software abstraction for an input or output medium of communication. Sockets allow communication between processes that lie on the same machine, or on different machines working in diverse environment and even across different continents. A socket is the most vital and fundamental entity. Sockets are the end-point of a two-way communication link. An endpoint is a combination of IP address and the port number. For Client-Server communication, Sockets are to be configured at the two ends to initiate a connection, Listen for incoming messages Send the responses at both ends .Establishing a bidirectional communication. **Datagram Socket** A datagram is an independent, self-contained piece of information		4	L2	4	1	2.4.1

	sent over a network whose arrival, arrival time, and content are not guaranteed. A datagram socket uses User Datagram Protocol (UDP) to facilitate the sending of datagrams (self-contained pieces of information) in an unreliable manner. Unreliable means that information sent via datagrams isn't guaranteed to make it to its destination. **Stream Socket:** A stream socket, or connected socket, is a socket through which data can be transmitted continuously. A stream socket is more akin to a live network, in which the communication link is continuously active. A stream socket is a "connected" socket through which data is transferred continuously. sock_obj = socket.socket(socket_family, socket_type, protocol=0) socket_family: - Defines family of protocols used as transport mechanism. Either AF_UNIX, or AF_INET (IP version 4 or IPv4). socket_type: Defines the types of communication between the two end-points. SOCK_STREAM (for connection-oriented protocols, e.g., TCP), or SOCK_DGRAM (for connectionless protocols e.g. UDP). protocol: We typically leave this field or set this field to zero. Example: #Socket client example in python import socket #create an AF_INET, STREAM socket (TCP) s = socket.socket(socket.AF_INET, socket.SOCK_STREAM) print 'Socket Created'					
6	**What is Automata? Give examples.** **<u>Ans:</u>** Automata theory is the study of abstract computational devices Abstract devices are (simplified) models of real computations Computations happen everywhere: On your laptop, on your cell phone, in nature, … finite automata- Devices with a finite amount of memory. Used to model “small” computers. push-downautomata Devices with infinite memory that can be accessed in a restricted w ay. Used to model parsers, etc.	4	L1	5	1	2.1.2

	Turing Machines-Devices with infinite memory. Used to model any computer. A common way to talk about words, number, pairs of words, etc. is by representing them as strings. To define strings, we start with an alphabet **An alphabet is a finite set of symbols.** Σ1 = {a, b, c, d, …, z}: the set of letters in English Σ2 = {0, 1, …, 9}: the set of (base 10) digits Σ3 = {a, b, …, z, #}: the set of letters plus the special symbol # Σ4 = {(,)}: the set of open and closed brackets A string over alphabet Σ is a finite sequence of symbols in Σ. The empty string will be denoted by ∈ abfbz is a string over Σ1 = {a, b, c, d, …, z} 9021 is a string over Σ2 = {0, 1, …, 9} ab#bc is a string over Σ3 = {a, b, …, z, #}))()(() is a string over Σ4 = {(,)} Languages can be used to describe problems with "yes/no" answers, for example: L1 = The set of all strings over Σ1 that contain the substring "SRM" L2 = The set of all strings over Σ2 that are divisible by 7 = {7, 14, 21, …} L3 = The set of all strings of the form s#s where s is any string over {a, b, …, z} L4 = The set of all strings over Σ4 where every (can be matched with a subsequent)					
7	**Find the out the geometry managers for the following options and give its syntax:** **a. I have a set of radio widgets and want place it next to each other along horizontal axis.** **b. I have a button widget which I want to place it on specify coordinates denoting the screen.** **<u>Ans:</u>** **Grid** rb1= tk. Radiobutton(root, text="Option 2", variable=var,	4	L2	5	2	2.4.1

	value=1) rb1.grid(row=0,column =0) rb1.grid(row=0,column=1) b1= tk.Button(root, text="Submit", Command= root.destroy) b1.place(x=50,y=50)					
	Part - B **(3 x10 = 30 Marks)** **Instructions: Answer all the Questions**					
8 (a)	**(i) What is the difference between the paint() and repaint() methods in a Java Applet?** **Ans:** **paint():** This method holds instructions to paint this component. In Java Swing, we can change the paintComponent() method instead of paint() method as paint calls paintBorder(), paintComponent() and paintChildren() m ethods. We cannot call this method directly instead we can call repaint(). **repaint():** This method cannot be overridden. It controls the update() -> paint() cycle. We can call this method to get a component to repaint itself. If we have done anything to change the look of the component but not the size then we can call this method.	4	L1	3	2	1.4.1 2.4.1

	(ii) Write a java program that implements a multi-thread application that has three threads. First thread generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number.
	Ans:
	import java.util.Random;
	class Square extends Thread {
	int x;
	Square(int n) {
	x = n;
	}
	public void run() {
	int sqr = x * x;
	System.out.println("Square of " + x + " = " + sqr);
		6	L2	3	2	

```
}
}
class Cube extends Thread {
int x;
Cube(int n) {
x = n;
}
public void run() {
int cub = x * x * x;
System.out.println("Cube of " + x + " = " + cub);
}
}
class Rnumber extends Thread {
public void run() {
Random random = new Random();
for (int i = 0; i < 5; i++) {
int randomInteger = random.nextInt(10);
System.out.println("Random Integer generated : " + randomInteger);
Square s = new Square(randomInteger);
s.start();
Cube c = new Cube(randomInteger);
c.start();
try {
Thread.sleep(1000);
} catch (InterruptedException ex) {
System.out.println(ex);
}
}
}
}
```

	public class ThreadP {
	public static void main(String[] args) {
	Rnumber n = new Rnumber();
	n.start();
	}
	}
	OUTPUT:
	Random Integer generated : 9
	Square of 9 = 81
	Cube of 9 = 729
	Random Integer generated : 8
	Square of 8 = 64
	Cube of 8 = 512
	Random Integer generated : 1
	Square of 1 = 1
	Cube of 1 = 1
	Random Integer generated : 3
	Square of 3 = 9
	Cube of 3 = 27
	Random Integer generated : 8
	Square of 8 = 64
	Cube of 8 = 512					

	(OR)					
8 (b)	**(i) Discuss the five steps for connecting JDBC.** **Ans:** JDBC stands for Java Database Connectivity. JDBC is a Java API to connect and execute the query with the database. It is a part of JavaSE (Java Standard Edition). JDBC API uses JDBC drivers to connect with the database. JDBC API to access tabular data stored in any relational database. By the help of JDBC API, we can save, update, delete and fetch data from the database. There are 6 basic steps to connect with JDBC.	4 6	L1 L3	3 3	2 2	1.4.1 2.4.2



Note: Explain 6 steps in details

(ii) Create two objects threadobj1 and threadobj2 for the class UserThreadPriority. Set the name of threadobj1 as "ThreadA" and threadobj2 as "ThreadB". Get a String and a Character from the user and assign into UserThreadPriority class variable k and c respectively. Call the start() method for the thread objects threadobj1 and threadobj2.

Ans:

```
import java.util.Scanner;

public class UserThreadPriority {

private String k;

private char c;

private Thread thread;

public UserThreadPriority(String k, char c) {

this.k = k;

this.c = c;

this.thread = new Thread(this);

}

public void run() {

System.out.println(&quot;Thread name: &quot; + thread.getName());

System.out.println(&quot;String: &quot; + k);

System.out.println(&quot;Character: &quot; + c);

}

public static void main(String[] args) {

Scanner scanner = new Scanner(System.in);

System.out.println(&quot;Enter a string: &quot;);
```

	String k = scanner.nextLine();
	System.out.println("Enter a character: ");
	char c = scanner.next().charAt(0);
	// Create two objects of UserThreadPriority
	UserThreadPriority threadobj1 = new UserThreadPriority(k, c);
	UserThreadPriority threadobj2 = new UserThreadPriority(k, c);
	// Set the name of threadobj1
	threadobj1.thread.setName("ThreadA");
	// Set the name of threadobj2
	threadobj2.thread.setName("ThreadB");
	// Start the two threads
	threadobj1.thread.start();
	threadobj2.thread.start();
	}
	}					

9 (a) OR	**(i) Explain Parallel Programming and illustrate parallelism using a multithreaded application in python.** **<u>Ans:</u>** Multitasking, in general, is the capability of performing multiple tasks simultaneously. Multithreading refers to concurrently executing multiple threads by rapidly switching the control of the CPU between threads (called context switching). Python Global Interpreter Lock limits one thread to run at a time even if the machine contains multiple processors. There are two types of multitasking in an OS: Process-based Thread-based import threading from threading import * def calculate_square(num):	4 6	L2 L3	4 4	2 2	1.4.1 2.4.1

print("Calculate the square of a given number")

for n in num:

print(f'The Square of {n} is:', n*n)

def calculate_cube(num):

print("Calculate the cube of a given number")

for n in num:

print(f'The Cube of {n} is:', n*n*n)

lst = [2, 4, 6, 8, 10]

thread1 = threading.Thread(target=calculate_square, args=(lst,))

thread2 = threading.Thread(target=calculate_cube, args=(lst,))

thread1.start()

thread2.start()

thread1.join()

thread2.join()

(ii) Create a TCP server that echoes back any message it receives from a client. Develop a Python client to send messages to the server and display the echoed response.

Ans:

TCP SERVER:

import socket

Define the server address and port

server_address = ('localhost', 12345)

Create a socket and bind it to the server address

server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

server_socket.bind(server_address)

Listen for incoming connecXons

server_socket.listen(1) print("TCP

server is running...")

while True:

Accept a connecXon

client_socket, client_address = server_socket.accept()

print(f"ConnecXon established with {client_address}")

try:

while True:

Receive data from the client

data = client_socket.recv(1024)

if not data: break

Echo the data back to the client

client_socket.sendall(data) except

ExcepXon as e: print(f"Error: {e}")

finally:

Close the client socket

client_socket.close()

TCP CLIENT: import socket

Define the server address and port

server_address = ('localhost', 12345)

Create a socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

try:

Connect to the server

client_socket.connect(server_address)

while True:

message = input("Enter a message (or 'exit' to quit): ")

if message == 'exit': break

Send the message to the server

client_socket.sendall(message.encode())

Receive and display the echoed response

response = client_socket.recv(1024)

	print(f'Server echoed: {response.decode()}")
	except ExcepXon as e: print(f"Error: {e}")
	finally:
	# Close the socket
	client_socket.close()					

	(OR)					
9 (b)	**(i) Illustrate in detail about functional programming paradigm with examples.** **Ans:** Functional programming is a programming paradigm in which it is tried to bind each and everything in pure mathematical functions. It is a declarative type of programming style that focuses on what to solve rather than how to solve. Functional programming paradigm is based on lambda calculus. Instead of statements, functional programming makes use of expressions. Unlike a statement, which is executed to assign variables, evaluation of an expression produces a value. Functional programming is a declarative paradigm because it relies on expressions and declarations rather than statements. Unlike procedures that depend on a local or global state, value outputs in FP depend only on the arguments passed to the function. Functional programming consists only of PURE functions. Concepts: Pure functions Recursion Referential transparency Functions are First-Class and can be Higher-Order Immutability **Note :** Explain all the concepts. (ii) Design a expense tracker that is used to keep track of the user's expenses. It has to give correct information to the users on their expenses and help them spend better using PySimpleGUI and PyData libraries.	4 6	L2 L3	4 4	2 2	1.4.1 2.4.2

	Install the PySimpleGUI package Create basic user interface elements with PySimpleGUI Create applications, such as a PySimpleGUI image viewer Integrate PySimpleGUI with Matplotlib Use computer vision in PySimpleGUI Package your PySimpleGUI application for Windows ● .Tkinter – To create the GUI. ● messagebox – To display a box containing information, warning, or error or asking a question. ● Ttk.Treeview – To display a table in the GUI window. ● Tkcalender.DateEntry – To enter a date. ● SQLite – To connect the Python script to the SQL database. Only the Tkinter library comes pre-installed with Python, so you need to run the following command in your terminal to install them: python -m pip install tkcalender sqlite 1. Importing the necessary modules and libraries 2. Connecting to the database and creating the GUI window 3. Creating the database and data manipulation functions **Importing the necessary modules and libraries:** import datetime import sqlite3 from tkcalendar import DateEntry from tkinter import * import tkinter.messagebox as mb import tkinter.ttk as ttk **Note:** Follow the steps and write the any example.					
10(a)	**(i) What is Symbolic Programming Paradigms? Explain.** **Ans:** Symbolic programming is a programming paradigm in which the program can manipulate its own formulas and program components as if they were plain data.Through	4 6	L1 L2	5 5	1 2	2.4.1 2.4.8

	symbolic programming, complex processes can be developed that build other more intricate processes by combining smaller units of logic or functionality. Symbolic programs can effectively modify themselves and appear to learn, which makes them better suited for applications such as artificial intelligence expert systems natural language processing and computer games. Languages that support symbolic programming include the Language LISP and Prolog Sympy-Symbolic Mathematics in Python SymPy is a Python library used to represent the symbolic mathematics SymPy is written entirely in Python and does not require any external libraries. To Install the sympy use command : pip install sympy Sympy support to perform the mathematical operations such as Algebraic manipulations Differentiation Integration Equation solving & Linear algebra Mathematical Operations Using Sympy: To Find the rational no : The Rational class represents a rational number as a pair of two Integers: the numerator and the denominator. Rational(5, 2) represents 5/2. Mathematical Operations Using Sympy: Some special constant e,pi,oo (Infinity) considered as symbol and evaluate the values. evalf evaluates the expression to a floating-point number. Calculus limit() -method **Note:** Explain in details above topics with example. **(ii) Construct a NFA and give the transmission table for the following regular expression for the given language = (a + ba)*bb (a + ab)*.**					

Ans:

To construct a Non-Deterministic Finite Automaton (NFA) for the regular expression `(a + ba)*bb (a + ab)*`, we can break down the expression into two parts: `(a + ba)*` and `(a + ab)*`.

Each part can be implemented as a separate NFA, and then we can combine them. Let's create NFAs for each part and then combine them:

1. NFA for `(a + ba)*`:

- State q0: Initial state
- State q1: Accepting state for `(a + ba)*`
- Transitions:
  - q0 on 'a' goes to q0 (loop for 'a')
  - q0 on 'b' goes to q1 (transition for 'ba')
  - q1 on 'a' goes to q0 (loop for 'a')
  - q1 on 'b' goes to q1 (stay for 'ba')

2. NFA for `(a + ab)*`:

- State q2: Initial state
- State q3: Accepting state for `(a + ab)*`
- Transitions:
  - q2 on 'a' goes to q3 (transition for 'a')
  - q2 on 'b' goes to q2 (loop for 'b')
  - q3 on 'a' goes to q3 (stay for 'ab')
  - q3 on 'b' goes to q3 (stay for 'ab')

Now, we can combine these two NFAs into a single NFA:

3. Combined NFA for `(a + ba)*bb (a + ab)*`:

- State q0: Initial state
- State q1: Accepting state for `(a + ba)*`
- State q2: Initial state for `(a + ab)*`
- State q3: Accepting state for `(a + ab)*`
- Transitions:
  - q0 on 'a' goes to q0 (loop for `(a + ba)*`)
  - q0 on 'b' goes to q1 (transition for `(a + ba)*`)
  - q1 on 'b' goes to q1 (stay for `(a + ba)*`)
  - q1 on 'a' goes to q0 (loop for `(a + ba)*`)
  - q2 on 'a' goes to q3 (transition for `(a + ab)*`)
  - q2 on 'b' goes to q2 (loop for `(a + ab)*`)
  - q3 on 'a' goes to q3 (stay for `(a + ab)*`)
  - q3 on 'b' goes to q3 (stay for `(a + ab)*`)

Transmission Table for the Combined NFA:

		State | 'a' | 'b' |
		|-------|------|------|
		| q0 | q0 | q1 |
		| q1 | q0 | q1 |
		| q2 | q3 | q2 |
		| q3 | q3 | q3 |
		
		This NFA recognizes strings that match the regular expression `(a + ba)*bb (a + ab)*`. It can be used to determine if a given string belongs to the specified language.
		
		In python, we implement this as follows -
		
		``
		# Define the NFA transitions using a dictionary
		nfa = {
		0: {'a': {0}, 'b': {1}},
		1: {'b': {1}},
		2: {'a': {3}, 'b': {2}},
		3: {'a': {3}},
		}
		# Define the set of accepting states
		accepting_states = {1, 3}
		
		# Function to simulate the NFA
		def simulate_nfa(input_string):
		current_states = {0}
		for char in input_string:
		next_states = set()
		for state in current_states:
		if char in nfa[state]:
		next_states.update(nfa[state][char])
		current_states = next_states
		
		return any(state in current_states for state in accepting_states)
		
		# Test the NFA with some strings
		test_strings = ["bb", "abb", "abab", "baa", "aabbaab"]
		for string in test_strings:
		if simulate_nfa(string):
		print(f"'{string}' is in the language")
		else:
		print(f"'{string}' is not in the language")
		``					

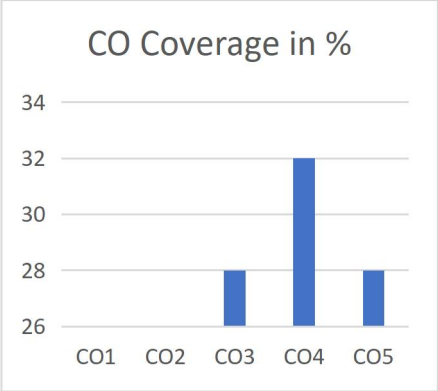
	(OR)					
10(b)	**(i) What functions are available for solving equations in SymPy.** **<u>Ans:</u>** SymPy is a Python library for symbolic mathematics, and it	4	L1	5	1	1.4.1

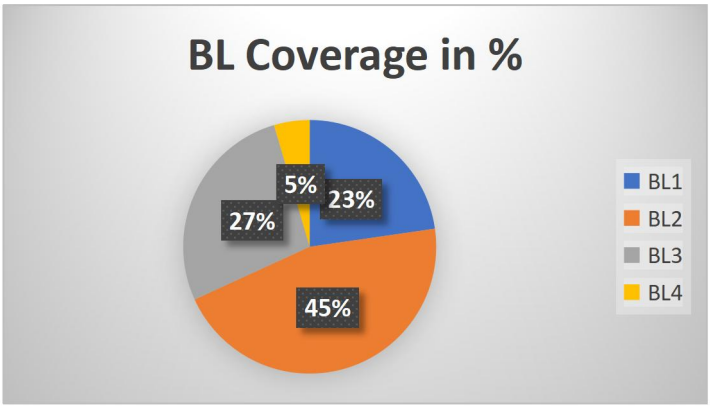
	provides a wide range of functions and tools for solving equations symbolically.
	Here are some of the key functions available for solving equations in SymPy:
	
	A. solve Function: The `solve` function is the primary tool for solving equations in SymPy. It can solve equations and systems of equations symbolically, including algebraic, transcendental, and differential equations.
	
	Example:
	from sympy import symbols, Eq, solve
	x, y = symbols('x y')
	equation = Eq(x**2 + y, 5)
	solutions = solve(equation, x)
	
	(ii) Create the calendar using Tkinter by showing data month and year with scroll down menu to fix the particular data month and year and press the click button to show the message of clicked date, month and year.
	
	Ans:
	
	rom tkinter import *
	
	import calendar
	
	from datetime import date
	
	def printCalendar():
	
	month = int(month_box.get())
	
	year = int(year_box.get())
	
	output_calendar = calendar.month(year, month)
	
	calendar_field.delete(1.0, 'end')
	
	calendar_field.insert('end', output_calendar)
	
	def reset():
	
	calendar_field.delete(1.0, 'end')
	
	month_var.set(current_month)
	
	year_var.set(current_year)
	
	month_box.config(textvariable=month_var)
	
	year_box.config(textvariable=year_var)
	
	def close():
	
	guiWindow.destroy()
	
	header_frame = Frame(guiWindow)
		6	L3	5	3	2.1.2

```
entry_frame = Frame(guiWindow)
result_frame = Frame(guiWindow)
button_frame = Frame(guiWindow)
header_frame.pack(expand=True, fill="both")
entry_frame.pack(expand=True, fill="both")
result_frame.pack(expand=True, fill="both")
button_frame.pack(expand=True, fill="both")
header_label = Label(header_frame, text="CALENDAR")
header_label.pack(expand=True, fill="both")
month_label = Label(entry_frame, text="Month:")
year_label = Label(entry_frame, text="Year:",
font=("arial", "20", "bold"), fg="#000000")
month_label.place(x=30, y=0)
year_label.place(x=275, y=0)
month_var = IntVar(entry_frame)
year_var = IntVar(entry_frame)
current_month = date.today().month
current_year = date.today().year
month_var.set(current_month)
year_var.set(current_year)
month_box = Spinbox(entry_frame, from_=1, to=12, width="10",
textvariable=month_var, font=('arial','15'))
year_box = Spinbox(entry_frame, from_=0000, to=3000,
width="10",
textvariable=year_var,font=('arial','15'))
month_box.place(x=130, y=5)
year_box.place(x=360, y=5)
if __name__ == "__main__":
guiWindow = Tk()
guiWindow.title("GUI Calendar")
guiWindow.geometry('500x550')
guiWindow.resizable(0, 0)
```

***Program Indicators are available separately for Computer Science and Engineering in AICTE examination reforms policy.**

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions





Approved by the Audit Professor/Course Coordinator